



MSc Data Analytics

Final Year Project

Impact of Data Balancing Techniques on Market Basket Analysis

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Abstract

This study investigates the impact of data balancing techniques on association rule mining in retail transaction data, with a particular focus on improving the detection of rules involving minority or rarely purchased items. Using the Online Retail dataset from the UCI Machine Learning Repository, four balancing methods random oversampling, random undersampling, hybrid balancing and the Synthetic Minority Oversampling Technique (SMOTE) were applied to the dataset and both the Apriori and FP-Growth algorithms were run on each version. Results were evaluated using support, confidence, lift, conviction, minority item rule coverage and execution time, with statistical significance assessed using the Mann-Whitney U test.

The study also employed the Gini coefficient as a measure of item frequency imbalance, confirming a baseline value of 0.196 which indicates moderate distributional skew in the dataset. This measure was used before and after each balancing technique to quantify the degree of improvement achieved. The Gini analysis revealed that the hybrid method produced the most balanced distribution with a coefficient of 0.1314, while SMOTE unexpectedly increased the Gini coefficient to 0.2589, further confirming its unsuitability for binary transaction data. Statistical significance of all results was confirmed using the Mann-Whitney U test at a 0.05 significance level.

The original imbalanced dataset produced 329 rules with a minority item coverage of 36.2%. Undersampling proved the most effective balancing method, increasing minority coverage to 52.2% whilst also achieving the highest average lift of 14.93. The hybrid approach was the second best with 49.5% minority coverage and 204 rules. Oversampling produced the most rules (469) but showed only modest improvement in minority coverage (39.9%). SMOTE produced the highest maximum lift (34.53) but the lowest minority coverage (14.6%), attributed to its unsuitability for binary transaction data. Both algorithms produced identical rule sets across all conditions, whilst FP-Growth was consistently faster up to 4.4 times faster than Apriori on the oversampled dataset. All differences in lift distributions were statistically significant. These findings provide practical guidance for retailers seeking to uncover non-obvious purchasing patterns involving less popular products.

Keywords: market basket analysis, association rule mining, data imbalance, Apriori, FP-Growth, oversampling, undersampling, SMOTE, minority items, retail data mining

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CHAPTER ONE: INTRODUCTION

1.1 Background

Retailers today collect enormous amounts of transactional data every day. Every purchase a customer makes leaves a record and when you combine millions of these records together, patterns start to emerge about what products tend to be bought together. Finding these patterns is the core purpose of market basket analysis, one of the most practical and well-established techniques in retail data mining. By discovering which products are frequently purchased together, retailers can make better decisions about product placement, promotions and personalised recommendations.

The most common algorithms used for this task are Apriori and FP-Growth, both of which were developed in the 1990s and early 2000s and have been widely studied ever since (Agrawal & Srikant, 1994; Han et al., 2000). These algorithms work by identifying itemsets that appear together frequently enough to exceed a minimum support threshold and then generating association rules from those itemsets. A typical rule might say something like customers who buy product A and product B also tend to buy product C in 70% of cases.

However, there is a well-known problem with this approach that does not get as much attention as it deserves. In any real retail dataset, some products are purchased far more frequently than others. A supermarket might sell bread and milk thousands of times a day, but a specialist kitchen gadget might only be sold a few times a week. When you run association rule mining on this kind of imbalanced data, the algorithms naturally focus on the popular items because they are the ones that meet the minimum support threshold. The rarer items, even if they have strong and consistent purchasing patterns often get ignored simply because they do not appear frequently enough.

This is a significant limitation. From a business perspective, discovering that people who buy bread also buy butter is not particularly useful most retailers already know this. What is far more valuable is discovering unexpected associations involving niche or less popular products, because these represent genuine insights that were not previously known. Unfortunately, standard association rule mining algorithms are not well suited to finding these patterns in imbalanced data.

1.2 Motivation for This Study

The motivation for this study came from recognising this gap between what association rule mining can theoretically do and what it actually does in practice when data is imbalanced. Data balancing techniques methods that modify the distribution of a dataset to reduce the dominance of majority items

have been widely studied in the context of classification problems. However, their application to association rule mining is much less explored.

It is not straightforward to simply apply balancing methods designed for classification to transaction data. In classification, you are trying to balance class labels. In association rule mining, you are working with binary transaction matrices where each row is a customer basket and each column is a product. Changing the distribution of this data changes the support, confidence and lift values of all rules, which can have complex and unpredictable effects on the output.

Despite this complexity, there is a genuine need to understand how different balancing approaches affect the results of association rule mining particularly whether they help reveal associations involving rarely purchased items. This study was designed to address that need directly, by systematically testing four different balancing methods and measuring their impact on both rule quality and minority item detection.

1.3 Aims and Objectives

This study investigates the impact of data balancing techniques on association rule mining, with a particular focus on whether these techniques can improve the detection of rules involving rarely purchased items in retail transaction data.

The specific objectives are:

- To analyse the degree of data imbalance present in the Online Retail dataset using appropriate measures including item frequency distribution and the Gini coefficient.
- To establish a baseline by applying Apriori and FP-Growth to the original imbalanced dataset and measuring rule quality using support, confidence, lift, conviction and minority item coverage.
- To apply four data balancing techniques random oversampling, random undersampling, hybrid balancing and SMOTE to the dataset and re-run both algorithms on each balanced version.
- To compare the results across all conditions in terms of rule quality metrics, minority item coverage and computational efficiency.
- To assess whether the differences in results are statistically significant using the Mann-Whitney U test.

1.4 Research Questions

This study is guided by five research questions:

- **RQ1:** How does data imbalance affect the quality of association rules generated from retail transaction data?
- **RQ2:** To what extent do data balancing techniques improve the detection of rare item associations in market basket analysis?
- **RQ3:** How do Apriori and FP-Growth algorithms compare in terms of rule quality and computational efficiency when applied to imbalanced and balanced datasets?
- **RQ4:** What is the effect of data balancing techniques on the statistical significance of association rules, as measured by metrics such as lift, conviction and Zhang's metric?
- **RQ5:** To what extent does the choice of data balancing technique affect the Gini coefficient of item frequency distribution in retail transaction datasets?

1.5 Dataset

The dataset used in this study is the Online Retail dataset, obtained from the UCI Machine Learning Repository. It contains over 1,067,371 transaction records from a UK-based online retailer covering the years 2009 to 2011.

The dataset includes fields such as invoice number, product description, quantity, price, customer ID and country of transaction. After preprocessing which involved removing cancelled transactions, missing values and non-UK records the cleaned dataset contained 27,738 transactions covering 150 of the most frequently purchased products. This dataset was chosen because it is a widely used benchmark in retail data mining research and because it naturally exhibits the kind of item frequency imbalance that this study aims to address.

1.6 Scope and Limitations

This study focuses specifically on market basket analysis using association rule mining. It does not cover other data mining tasks such as classification, clustering or regression. The experiments were conducted on a single retail dataset the Online Retail dataset from the UCI Machine Learning Repository. While this is a well-established benchmark dataset, the findings may not generalise to all

types of retail environments or product categories. Different industries, product ranges or transaction volumes could produce different results.

The study uses fixed parameter settings throughout all experiments minimum support of 0.01, minimum confidence of 0.30 and minimum lift of 1.0. A full sensitivity analysis across different parameter values was not conducted, which means it is possible that different thresholds could produce different results.

Additionally, SMOTE was applied in its standard form, which was originally designed for continuous numerical data rather than binary transaction matrices. This is acknowledged as a limitation of the current study. These limitations are discussed further in Chapter Five.

1.7 Structure of the Dissertation

The dissertation is structured across six chapters covering introduction, literature review, methodology, results, discussion and conclusion respectively, followed by a references section and appendices.



Figure 1.1: Project Lifecycle

1.8 Key Contributions

This research makes several important contributions to the field of retail data mining and association rule analysis:

- A comprehensive analysis of data imbalance in transaction datasets using statistical measures such as median frequency and Gini coefficient.
- Implementation and comparison of multiple data balancing techniques, including random oversampling, random undersampling, hybrid sampling and SMOTE.
- Evaluation of association rule mining algorithms (Apriori and FP-Growth) under different data balancing conditions.
- Introduction of minority item coverage as an evaluation metric to assess the representation of rare items.
- Statistical validation of results using the Mann–Whitney U test to determine the significance of improvements.

These contributions provide deeper insights into how data imbalance affects rule mining performance and how it can be effectively addressed.

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

Market basket analysis has been one of the most widely used techniques in retail data mining since it was first introduced by Agrawal & Srikant (1994). It is an important data mining technique used to identify relationships between items purchased together in transactional datasets. It is widely applied in retail environments to improve recommendation systems, product placement and business decision-making. One persistent challenge, however, is data imbalance. In any real retail dataset, some products sell far more than others. This imbalance can reduce the effectiveness of association rule mining algorithms and limit the discovery of meaningful patterns.

This chapter reviews research on market basket analysis, association rule mining algorithms and data balancing techniques relevant to this study. It also examines previous studies that address challenges associated with imbalanced datasets. Finally, the research gap that motivates this project is identified.

2.2 Market Basket Analysis

Market basket analysis works by scanning transaction records to find items that appear together more often than would be expected by chance. The output is a set of association rules in the form $\{A\} \rightarrow \{B\}$, which can be read as: "customers who buy A also tend to buy B." Each rule is evaluated using three standard measures: support, confidence and lift (Agrawal & Srikant, 1994).

Support tells you how often a combination of items appears across all transactions. Confidence measures the probability that a customer buys B given that they have already bought A. Lift puts confidence in context by comparing it against what you would expect if A and B were bought independently a lift value above 1 means the items are genuinely associated, while a value below 1 suggests they are actually less likely to appear together than by chance (Tan et al., 2005).

Data mining textbooks such as Witten et al. (2016) describe market basket analysis as one of the most practically valuable applications of association rule mining in commercial settings. Kulkarni et al. (2023) further demonstrated that market basket analysis remains a foundational technique in modern recommender systems, particularly in e-commerce retail environments.



Figure 2.1: Market basket analysis showing relationships between items.

However, these metrics generally assume that item distribution is balanced which is not true in real retail datasets. In highly imbalanced datasets, they tend to favour frequent items, reducing the visibility of meaningful associations involving low-frequency products.

2.3 Association Rule Mining Algorithms

2.3.1 Apriori Algorithm

Apriori was one of the first algorithms developed for association rule mining and remains widely used today (Agrawal & Srikant, 1994). It works by starting with individual items and checking which ones meet the minimum support threshold. It then combines those frequent single items into pairs, checks which pairs are frequent, then triples and so on. Any itemset that falls below the support threshold is pruned immediately, which avoids having to check all possible combinations.

Although Apriori is effective in reducing the search space, its reliance on minimum support thresholds makes it inherently biased against rare itemsets, which are often eliminated early in the pruning process.

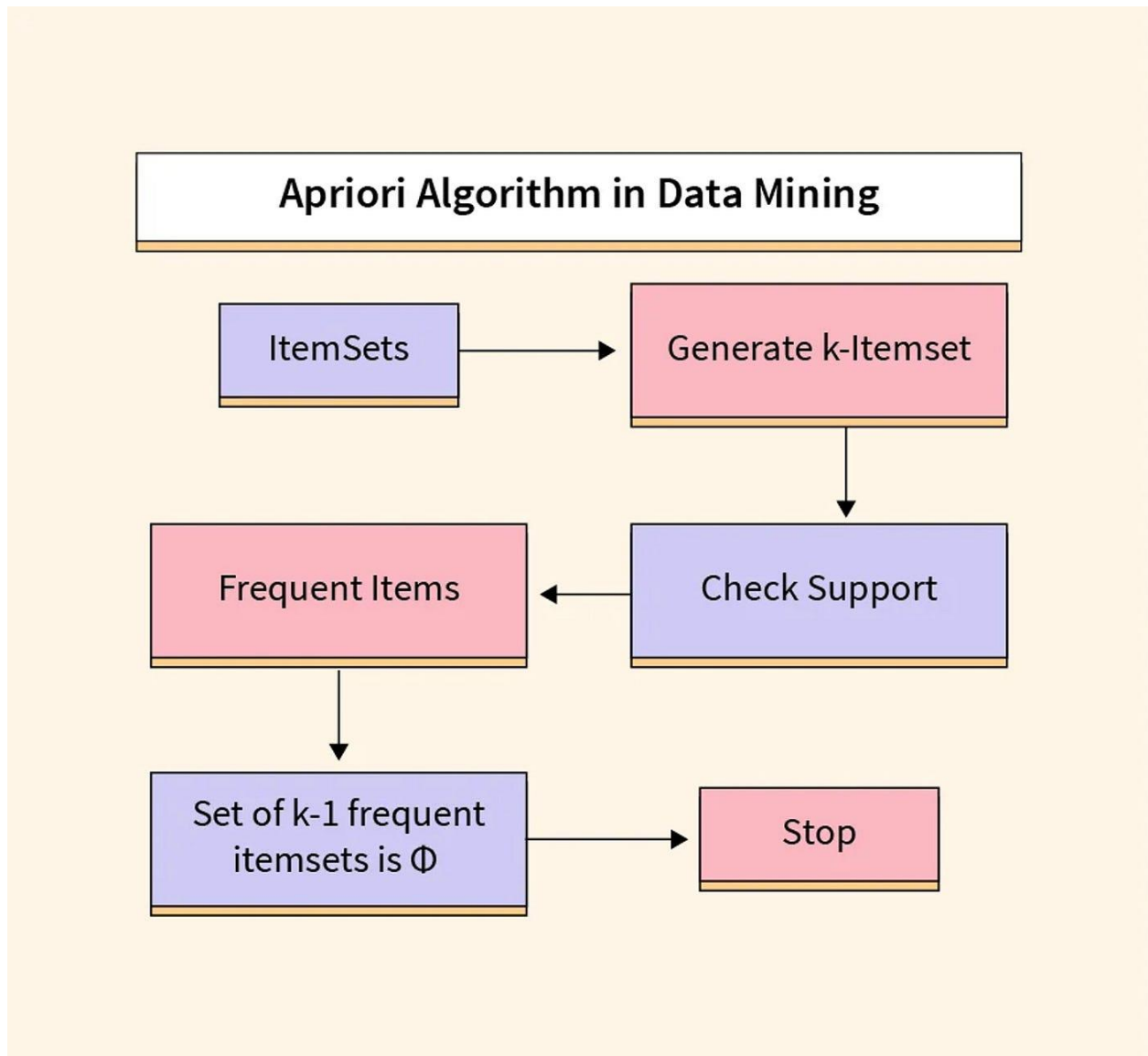


Figure 2.2: Working process of Apriori algorithm

2.3.2 FP-Growth Algorithm

FP-Growth was developed by Han et al. (2000) to overcome the multiple-scan limitation of Apriori. Instead of generating candidate itemsets explicitly, it compresses the entire transaction database into a tree structure called an FP-tree, which only requires two passes through the data. Frequent patterns are then extracted from this tree using a divide-and-conquer approach, without ever generating candidate itemsets in the traditional sense.

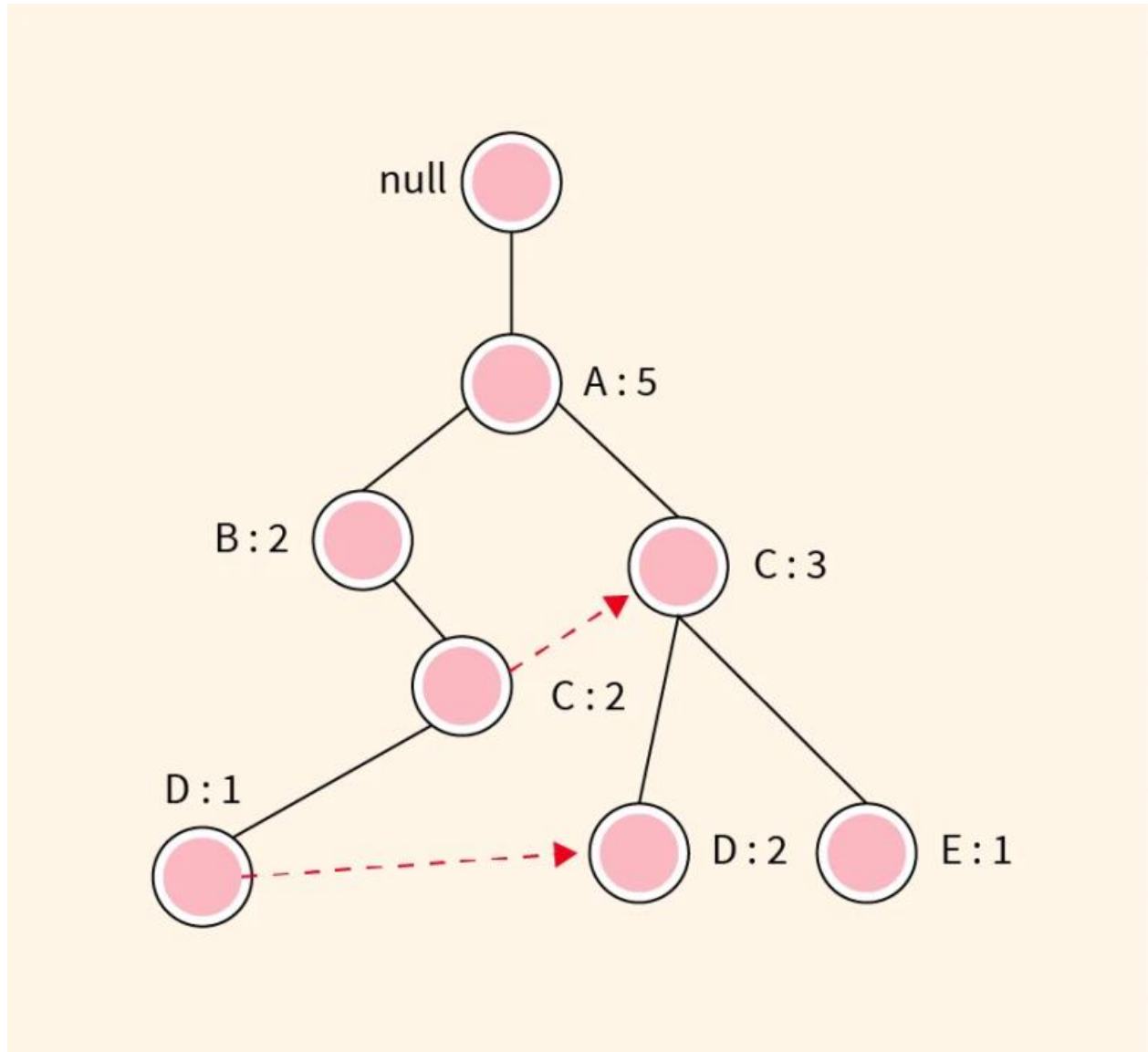


Figure 2.3: FP-tree structure used in FP-Growth algorithm.

FP-Growth is more efficient because it reduces database scans but it still has limitations when dealing with rare items.

2.3.3 Comparing Apriori and FP-Growth on Imbalanced Data

Both algorithms miss minority item patterns but through the different mechanisms. Luna et al. (2019) compared them across datasets with varying frequency skew and found that FP-Growth tends to generate fewer rules but with higher average confidence while Apriori generates more rules with more variation in confidence. This is important for this project because balancing may increase the frequency of rare items.

As a result, Apriori and FP-Growth may behave differently. Heaton (2017) compared dataset characteristics that favour each algorithm and found that FP-Growth performs better on denser datasets while Apriori is more efficient on sparse data, which is relevant given the 3.89% density of the dataset used in this study. More recently, Nande et al. (2025) evaluated the performance of Apriori, Apriori-TID and FP-Growth across diverse datasets and confirmed that FP-Growth consistently outperforms Apriori in terms of execution time across different data conditions. Sreelakshmi et al. (2025) proposed a hybrid Apriori-FP-Growth approach for retail decision-making and demonstrated that combining both algorithms can improve rule discovery efficiency in large-scale transaction datasets.

Therefore, while Apriori may generate a larger number of rules, FP-Growth tends to produce more concise and higher-confidence rules. However, both algorithms remain limited in their ability to capture minority-item associations in imbalanced datasets, reinforcing the need for data balancing approaches.

2.4 Data Imbalance in Retail Datasets

Data imbalance in retail transaction data is not unusual it is actually the norm. In most real retail datasets a small number of products account for the vast majority of transactions, while most other products are bought only occasionally. Krawczyk (2016) notes that in many datasets the top 20% of items can account for up to 80% of all item appearances, which creates a severe mismatch with minimum support thresholds that treat all items equally.

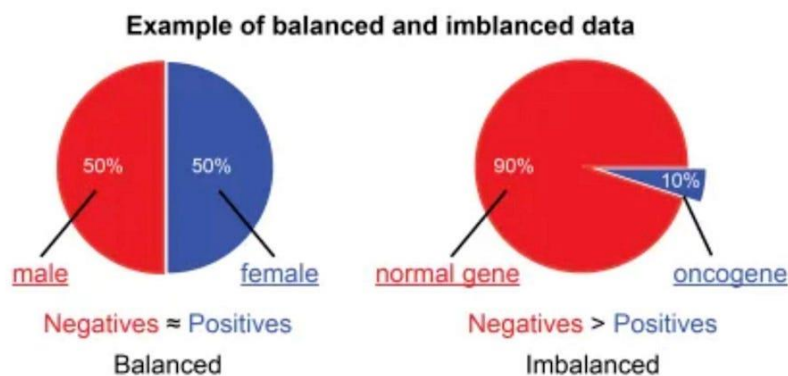


Figure 2.4: Example of imbalanced dataset distribution.

The practical consequence is that rules derived from imbalanced data mostly reflect popular products, while potentially valuable patterns involving niche or less frequently bought items are systematically missed. Garcia et al. (2012) also pointed out that datasets collected over short time windows make this worse, because seasonal or newer products have had less time to accumulate transactions even if their co-occurrence patterns are consistent. This is directly relevant here, since the Online Retail dataset includes products with very uneven purchase frequencies. Thabtah et al. (2020) experimentally evaluated data imbalance in classification tasks and confirmed that imbalanced distributions consistently reduce the ability of standard algorithms to detect minority patterns, a finding that applies equally to association rule mining. Hassan & Hassan (2024) further highlighted in their systematic review of machine learning in retail that data imbalance remains one of the most persistent and underaddressed challenges in retail analytics. This imbalance does not only affect data distribution but also directly influences the rule generation process, as minimum support thresholds disproportionately favour high-frequency items.

2.5 Data Balancing Techniques

Data balancing methods aim to reduce the skew in a dataset's distribution. Oversampling increases the representation of minority instances and undersampling reduces the majority instances. Random versions of both are simple but have drawbacks random undersampling risks losing useful information, while random oversampling just duplicates existing instances and can cause overfitting (Fernández et al., 2018). One widely used oversampling technique is the Synthetic Minority Over-sampling Technique (SMOTE), introduced by Chawla et al. (2002), which generates synthetic samples based on existing minority data points.

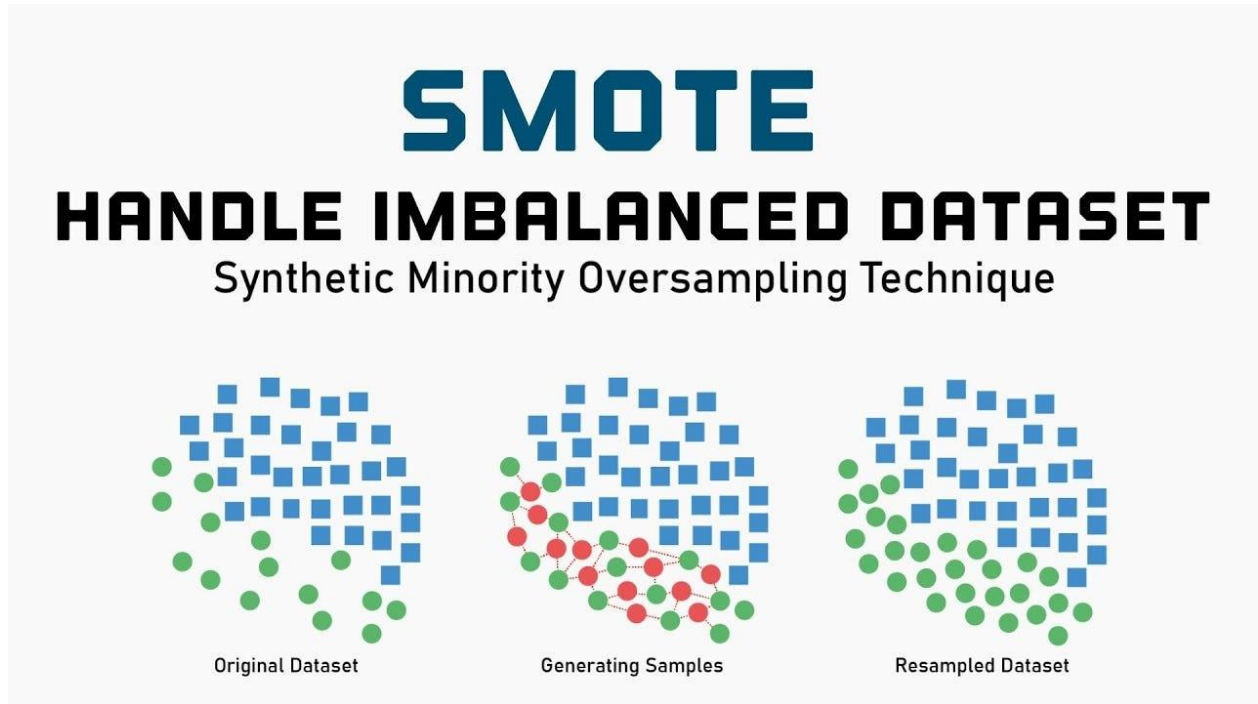


Figure 2.5: SMOTE technique for data balancing.

Applying balancing to association rule mining is more complicated than applying it to classification. In classification, SMOTE generates synthetic instances with class labels. In association rule mining, the equivalent means generating synthetic transactions that include minority items, which raises their support values and changes lift and confidence for all rules involving them. Yang et al. (2024) reviewed oversampling techniques across multi-class imbalanced datasets and noted that SMOTE's interpolation approach is most effective when the minority class instances form coherent clusters in feature space, a condition that is not guaranteed in binary transaction data.

Liu et al. (1999) showed that these distributional shifts can produce counterintuitive effects on rule metrics and Zaki (2000) noted that the minimum support threshold effectively assumes something about how items are distributed an assumption that needs careful re-evaluation after balancing. Elreedy & Atiya (2019) conducted a comprehensive analysis of SMOTE and confirmed that its effectiveness diminishes significantly when applied to non-continuous data, which directly supports the finding in this study that SMOTE underperforms on binary transaction matrices. Kaur & Gosain (2018) compared the behaviour of oversampling and undersampling approaches and found that combining both methods generally outperforms either technique applied in isolation, particularly when noise is present in the data. Mohammed et al. (2020) similarly evaluated oversampling and undersampling across multiple machine learning tasks and concluded that the effectiveness of each technique is highly

dependent on the underlying data distribution. Unlike classification problems, applying balancing techniques in association rule mining changes the transaction data rather than class labels. This directly impacts support, confidence and lift values, potentially changing the significance and ranking of generated rules.

2.6 Related Research Studies

More recent work by Luna et al. (2019) provided a comprehensive survey of frequent itemset mining algorithms and compared their performance across datasets with varying levels of frequency skew. Their findings confirmed that both Apriori and FP-Growth struggle to detect minority item associations when imbalance is severe, as the minimum support threshold consistently eliminates infrequent items before any rules can be generated. Similarly, Krawczyk (2016) conducted an extensive review of learning from imbalanced data and highlighted that minimum support thresholds systematically disadvantage rare items regardless of which algorithm is used. Garcia et al. (2012) reinforced this view, noting that datasets collected over short time windows make the imbalance problem worse because newer or seasonal products have had less time to accumulate transaction counts even when their co-occurrence patterns may be genuine and consistent. He & Garcia (2009) similarly observed that class imbalance in machine learning broadly leads to biased models that over-represent majority patterns, a finding that translates directly to the association rule mining context.

In terms of data balancing, Chawla et al. (2002) introduced SMOTE and demonstrated its effectiveness in classification tasks by generating synthetic minority class instances through interpolation between existing minority samples and their nearest neighbours. Their work became one of the most influential contributions to the imbalanced learning literature. Fernandez et al. (2018) later provided a comprehensive review of balancing techniques across multiple domains, evaluating both oversampling and undersampling strategies. However, the majority of this body of work focuses on classification problems rather than association rule mining. Liu et al. (1999) explored the use of multiple minimum support thresholds to give different items different frequency requirements, which is one approach to addressing item imbalance without resampling the data. Zaki (2000) noted that applying balancing to association rule mining changes the support and confidence values of rules in ways that are not always straightforward to predict.

However, neither of these studies provided a systematic comparison of multiple balancing methods applied to the same imbalanced retail dataset using both algorithms simultaneously. Wahidi & Ismailova (2024) applied association rule mining to e-commerce retail data and demonstrated the practical relevance of these techniques in online retail environments similar to the dataset used in this study. Kholod & Mokrenko (2024) applied rule-based algorithms and data mining techniques to market basket analysis and highlighted the ongoing challenge of handling imbalanced transaction frequencies in retail datasets.

Table 2.1: Summary of Related Research Studies

Author(s)	Year	Topic	Method	Key Finding
Agrawal & Srikant	1994	Association rule mining	Apriori algorithm	Introduced Apriori for frequent itemset mining
Han et al.	2000	Association rule mining	FP-Growth algorithm	FP-Growth faster than Apriori
Chawla et al.	2002	Imbalanced data	SMOTE	SMOTE effective for classification tasks
Liu et al.	1999	Imbalanced data	Multiple support thresholds	Minority items need lower thresholds
Krawczyk	2016	Imbalanced learning	Survey	Minimum support biased toward majority items
Fernandez et al.	2018	Imbalanced data	Survey	Undersampling effective for reducing majority dominance
Luna et al.	2019	Frequent itemset mining	Review	Gap exists in balancing for association rule mining

Elreedy & Atiya	2019	SMOTE analysis	Review	SMOTE less effective for non-continuous data
Witten et al.	2016	Data mining	Textbook	MBA is most valuable application of ARM
Kulkarni et al.	2023	Market basket	Recommender systems	MBA foundational in e-commerce
Heaton	2017	Algorithm comparison	Dataset characteristics	FP-Growth better on dense data
Nande et al.	2025	Algorithm evaluation	Comparison	FP-Growth consistently faster
Sreelakshmi et al.	2025	Retail decision-making	Hybrid ARM	Hybrid approach improves efficiency
Thabtah et al.	2020	Data imbalance	Experimental	Imbalance reduces minority detection
Hassan & Hassan	2024	Retail ML review	Systematic review	Imbalance underaddressed in retail
Kaur & Gosain	2018	Balancing techniques	Comparison	Combined methods outperform single
Mohammed et al.	2020	Oversampling/undersampling	Evaluation	Effectiveness depends on distribution
Yang et al.	2024	Oversampling review	Multi-class	SMOTE needs coherent clusters
Wahidi & Ismailova	2024	ARM in e-commerce	Implementation	Practical relevance confirmed

Kholod & Mokrenko	2024	Market basket	Rule-based	Imbalance challenge highlighted
Chen et al.	2012	Online retail dataset	RFM model	Dataset suitable for retail mining

2.7 Research Gap

Despite extensive research on association rule mining and data balancing independently, there is limited work that integrates both areas in a controlled and systematic way. The existing literature has established clearly that standard algorithms such as Apriori and FP-Growth are biased towards majority items in imbalanced datasets and that data balancing techniques can address distributional skew in classification contexts. However, no study has conducted a direct and controlled comparison of how oversampling, undersampling, hybrid balancing and SMOTE each affect the performance of both Apriori and FP-Growth on the same imbalanced retail dataset, using multiple rule quality metrics including minority item coverage as the primary evaluation criterion.

Most existing studies either focus on improving overall rule accuracy without considering minority items specifically or apply balancing techniques to classification problems without transferring those insights to association rule mining. Where balancing has been applied to transactional data, the evaluations have typically focused on a single balancing method, a single algorithm, or a single evaluation metric, making it difficult to draw comparative conclusions across methods.

Furthermore, the unique challenge of applying SMOTE to binary transaction data where synthetic interpolation produces non-binary values that must be rounded has not been systematically evaluated in the context of association rule mining. These gaps in the existing literature directly motivate the five research questions outlined in Chapter One, which form the basis of this study. By addressing all four balancing techniques on the same dataset with two algorithms and a consistent set of evaluation metrics, this study aims to provide a more complete and practically useful understanding of how balancing affects association rule quality in imbalanced retail data.

2.8 Summary

This chapter has reviewed the key literature underpinning this study across four areas: market basket analysis, association rule mining algorithms, data imbalance in retail datasets and data balancing techniques. The review confirms a genuine gap in the literature around the combined effect of balancing technique choice and algorithm choice on association rule quality in imbalanced retail datasets, which the following chapters address through a controlled experimental methodology.

CHAPTER THREE: METHODOLOGY

3.1 Research Philosophy

This study is grounded in a positivist research philosophy. Positivism holds that knowledge is derived from observable, measurable phenomena and that research findings should be based on empirical evidence rather than subjective interpretation. This philosophy is appropriate for the current study because the research questions are answered through quantitative measurement of association rule quality metrics support, confidence, lift, conviction and minority item coverage across controlled experimental conditions. The use of statistical testing through the Mann-Whitney U test further reflects the positivist commitment to objective, verifiable evidence. This approach is consistent with the broader tradition of experimental computer science research, in which conclusions are drawn from reproducible experiments conducted under controlled conditions rather than from qualitative observation or human judgement.

3.2 Research Approach

This study follows a quantitative and experimental research design. The primary objective is to investigate how different data balancing techniques affect the quality of association rules generated by Apriori and FP-Growth algorithms when applied to imbalanced retail transaction data. A quantitative approach was chosen because the research questions require measurable comparisons across multiple experimental conditions using defined evaluation metrics such as support, confidence, lift, conviction and minority item coverage. The experimental design involves applying four balancing techniques to the same dataset and running both algorithms under identical parameter settings across all conditions, allowing direct and controlled comparison of results. This approach is consistent with the methodology used in related studies such as Luna et al. (2019) and Fernandez et al. (2018), who also employed controlled experimental designs to compare algorithm performance across different data conditions. The study does not involve human participants and therefore does not raise ethical concerns beyond standard data handling practices.

This study follows the Cross-Industry Standard Process for Data Mining (CRISP-DM) framework, which provides a structured approach to data mining projects. The CRISP-DM framework consists of six phases: Business Understanding, Data Understanding, Data Preparation, Modelling, Evaluation and Deployment. In this study, the Business Understanding phase involved identifying the problem of

data imbalance in retail transaction data and defining the research questions. The Data Understanding phase involved exploring the UCI Online Retail dataset and measuring item frequency imbalance using the Gini coefficient. The Data Preparation phase involved preprocessing the dataset and applying four data balancing techniques. The Modelling phase involved applying both Apriori and FP-Growth algorithms across all five dataset versions. The Evaluation phase involved comparing results using rule quality metrics, minority item coverage and statistical significance testing. As this is an academic study, the Deployment phase is not applicable, however practical recommendations are provided in Chapter Six for retailers wishing to apply these findings.

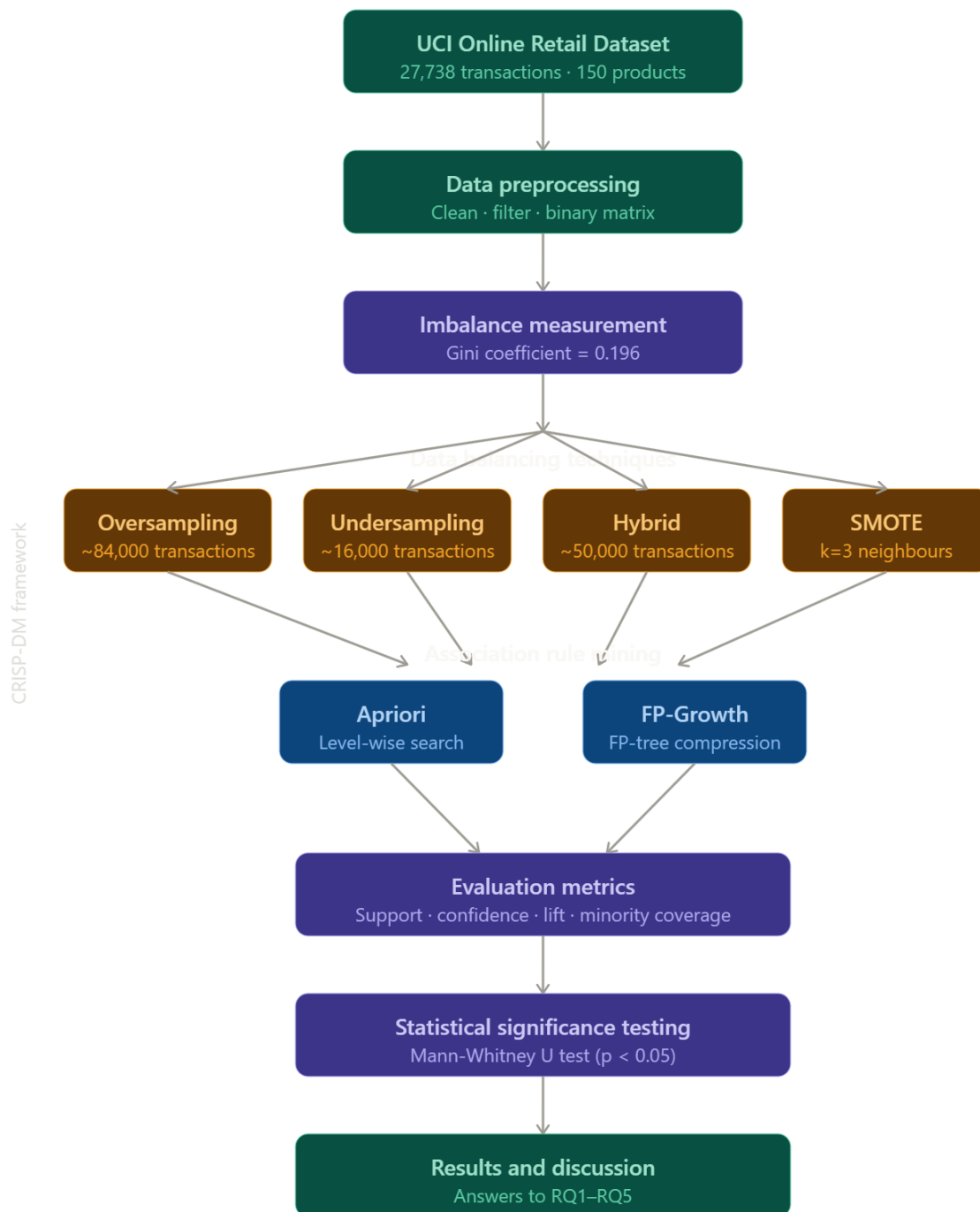


Figure 3.1: Research Framework and Methodology Overview

Figure 3.1 provides an overview of the complete research framework adopted in this study, illustrating the flow from data collection through preprocessing, balancing, algorithm application, evaluation and results.

3.3 Dataset Description

The dataset used in this study is the Online Retail dataset, obtained from the UCI Machine Learning Repository. It contains transactional records from a UK-based online retail store covering the period

from December 2009 to December 2011. The full dataset contains 1,067,371 rows across eight columns: Invoice, StockCode, Description, Quantity, InvoiceDate, Price, Customer ID and Country. Each row represents a single product within a transaction and multiple rows share the same Invoice number when a customer purchased more than one item in a single order. Chen et al. (2012) previously used this dataset for RFM model-based customer segmentation and confirmed its suitability for retail data mining research, further supporting its use as the benchmark dataset in this study.

This dataset was selected for three reasons. First, it is a well-established benchmark dataset widely used in retail data mining research. Second, it represents genuine real-world purchasing behaviour with natural item frequency imbalance. Third, it is sufficiently large to support meaningful statistical analysis after preprocessing. After applying the preprocessing steps described in Section 3.4, the final dataset used in the experiments contained 27,738 transactions covering the top 150 most frequently purchased products. The resulting binary transaction matrix had a density of 3.89%, confirming that the data is highly sparse, which is typical of real retail transaction datasets.

3.4 Data Preprocessing

Data preprocessing was carried out in several steps to ensure the dataset was clean and suitable for association rule mining. First, all rows with missing values in the Description or Customer ID fields were removed, as incomplete records cannot be meaningfully assigned to a transaction. Second, cancelled transactions were identified by invoice numbers beginning with the letter 'C' and removed, since these represent returns rather than purchases. Third, records with negative quantities or zero unit prices were excluded, as these indicate adjustments or errors rather than genuine sales. Fourth, only transactions from the United Kingdom were retained to ensure a consistent purchasing context, since international orders represent a small proportion of the data and may reflect different buying patterns.

After these cleaning steps, the data was filtered to retain only the top 150 most frequently purchased products, as using all products would create an extremely sparse matrix that is computationally impractical for association rule mining. Each transaction was then converted into a binary matrix format where each row represents a single transaction and each column represents one of the 150 products, with a value of 1 indicating that the product was purchased in that transaction and 0 indicating it was not. This binary matrix served as the input for all experiments. The degree of item frequency imbalance in the cleaned dataset was measured using the Gini coefficient, which returned a value of 0.196, confirming moderate distributional skew. The median item frequency across all 150 products was 919 transactions and products appearing in fewer transactions than this median were classified as

minority items for the purposes of evaluation, following the operational definition adopted in this study.

3.5 Algorithms Used

Two association rule mining algorithms were implemented in this study: Apriori and FP-Growth. Apriori, introduced by Agrawal & Srikant (1994), generates frequent itemsets through an iterative level-wise search. It begins with individual items, checks which meet the minimum support threshold, then progressively builds larger itemsets by combining frequent smaller ones. Any itemset that falls below the support threshold is pruned immediately, which reduces the search space. However, Apriori requires multiple passes through the database one for each itemset size which becomes computationally expensive on large datasets.

FP-Growth, introduced by Han et al. (2000), addresses this limitation by compressing the entire transaction database into a compact tree structure called an FP-tree in just two passes through the data. Frequent patterns are then extracted directly from this tree using a divide-and-conquer approach, without generating explicit candidate itemsets. Both algorithms were implemented using the mlxtend library in Python and applied to all five dataset versions the original and the four balanced variants under identical parameter settings: minimum support of 0.01, minimum confidence of 0.30, minimum lift of 1.0 and maximum itemset length of 3. These thresholds were chosen based on standard practice in retail association rule mining research and were held constant across all conditions to ensure that any differences in results were attributable solely to the balancing technique rather than to parameter variation.

3.6 Data Balancing Techniques

Four data balancing techniques were applied to the original dataset to create four modified versions for comparison. Each technique was applied independently so that its individual effect could be assessed in isolation.

Random oversampling was applied by duplicating all transactions that contained at least one minority item three additional times. This increased the absolute frequency of minority item appearances in the

dataset, raising their support values and making them more likely to appear in generated rules. The oversampled dataset contained approximately 84,000 transactions.

Random undersampling was applied by identifying the top 15% of products by transaction frequency those appearing in more than the 85th percentile of item frequencies and randomly removing 40% of the transactions that contained these dominant products. This reduced the relative dominance of majority items without adding synthetic data. The undersampled dataset contained approximately 16,000 transactions.

Hybrid balancing combined both of the above approaches. Minority item transactions were duplicated three times and simultaneously 20% of transactions involving the most dominant products were removed. This aimed to compress the frequency gap from both ends. The hybrid dataset contained approximately 50,000 transactions.

SMOTE (Synthetic Minority Over-sampling Technique), introduced by Chawla et al. (2002), was applied using $k=3$ nearest neighbours to generate synthetic minority transactions. For each minority transaction, SMOTE identifies the three most similar minority transactions in the feature space and interpolates between them to create new synthetic instances. In this study, SMOTE was applied using the standard implementation, which was originally designed for continuous numerical data. The binary nature of the transaction matrix means that the interpolated values are non-binary decimals that must be rounded back to 0 or 1, which is a known limitation of applying standard SMOTE to transactional data. The degree of imbalance before and after each balancing method was quantified using the Gini coefficient to provide an objective measure of how much each technique reduced distributional skew.

To ensure reproducibility and consistency across experiments, all random sampling operations used in oversampling, undersampling and hybrid balancing were performed using a fixed random seed in Python.

3.7 Evaluation Metrics

The performance of the algorithms across all experimental conditions was evaluated using five rule quality metrics: support, confidence, lift, conviction and minority item rule coverage. Support measures the proportion of transactions in the dataset that contain all items in a given rule. It reflects how frequently a pattern occurs and determines whether a rule meets the minimum threshold for inclusion.

Confidence measures the conditional probability that a customer purchases the consequent item given that they have already purchased the antecedent. A higher confidence value indicates a more reliable rule.

Lift measures the strength of association between the antecedent and consequent relative to what would be expected if they were statistically independent. A lift value greater than 1 indicates a genuine positive association, while a value of exactly 1 indicates no relationship. Lift was used as the primary quality metric for comparing rule strength across conditions because it accounts for item frequency and is therefore more informative than support or confidence alone in imbalanced contexts.

Conviction measures the degree to which the rule would be incorrect if the association were random. A higher conviction value indicates a stronger directional dependency between the antecedent and the consequent and provides a complementary perspective to lift.

The minority item rule coverage was the primary measure of success in this study. It is defined as the percentage of generated rules that involve at least one item with a below-median transaction frequency across the 150 products studied. This metric directly reflects the extent to which each balancing method improves the algorithm's ability to detect associations involving rarely purchased products which is the central objective of this research.

In addition to rule quality metrics, the total number of rules generated under each condition and the execution time in seconds for each algorithm were recorded. Execution time was measured as the wall-clock time taken from the start of the frequent itemset mining step to the completion of rule generation. Statistical significance of differences in lift distributions between each balanced condition and the baseline was assessed using the Mann-Whitney U test at a significance level of 0.05. This non-parametric test was selected because it does not assume normal distribution of the data, which is appropriate given that lift values in association rule mining are typically skewed.

3.8 Experimental Procedure

The experiment was conducted in five stages. In the first stage, the original cleaned dataset was used to establish a baseline. Both Apriori and FP-Growth were applied to this unmodified dataset under the standard parameter settings and the resulting rules were recorded along with all evaluation metrics. In the second stage, each of the four balancing techniques was applied to the original dataset separately

to create four modified datasets: oversampled, undersampled, hybrid and SMOTE-balanced. In the third stage, both algorithms were applied to each of the four balanced datasets using identical parameter settings to those used in the baseline experiment. In the fourth stage, the results from all ten runs two algorithms across five dataset versions were compiled into a summary table for comparison. In the fifth stage, the Mann-Whitney U test was applied to the lift distributions from each balanced condition against the baseline to determine whether the observed differences in rule quality were statistically significant at a significance level of 0.05. All experiments were implemented in Python using the mlxtend, pandas, numpy, scipy and matplotlib libraries and all outputs including rule sets, summary statistics and figures were saved for analysis and inclusion in this report.

3.9 Ethical, Legal and Professional Considerations

This study uses a publicly available dataset obtained from the UCI Machine Learning Repository (Chen, 2015). No personal or identifiable information was collected or processed. The dataset contains only anonymised transaction records with no sensitive personal information. All analysis was conducted purely for academic purposes in accordance with the University of Portsmouth Ethics Policy. Full ethical approval was obtained prior to commencing this study, as documented in Appendix A.

From a legal perspective, the dataset is freely available for academic use and no copyright or data protection issues arise from its use. All software tools used in this study, including Python and the mlxtend library, are open source and freely available under standard academic licences.

Professionally, all sources, datasets, tools and references used in this study have been properly cited and acknowledged throughout the dissertation in accordance with the University of Portsmouth academic integrity guidelines.

CHAPTER FOUR: RESULTS

4.1 Chapter Overview

This chapter presents the experimental results across all five dataset conditions using both Apriori and FP-Growth algorithms.

4.2 Dataset Characteristics and Imbalance

The online retail dataset was obtained from the UCI Machine Learning Repository. It contains transaction records from a UK based online retailer between 2009 and 2011. After cleaning the data which involved removing cancelled orders, missing customer IDs, negative quantities and transactions from outside the UK and the final dataset contained 27,738 transactions. The top 150 most frequently purchased products were selected to build the binary transaction matrix used in the experiments.

The transaction matrix had a density of only 3.89%, meaning that on average customers only bought a small number of products out of the 150 available in any given transaction. This kind of sparsity is quite normal in retail data, since most customers will only purchase a handful of items at a time. Looking at how frequently each product appeared in transactions, there was a very obvious imbalance. As shown in Figure 4.1, the most popular product WHITE HANGING HEART T-LIGHT HOLDER appeared in over 4,600 transactions, while many products appeared fewer than 50 times. This gives an imbalance ratio of over 90:1 between the most and least purchased items.

To measure this imbalance properly, the Gini coefficient was calculated across all item frequencies. The result was 0.196, which indicates a substantial level of inequality in how often products appear. A Gini value of 0 would mean all products are purchased equally and a value of 1 would mean all purchases are concentrated in a single product. A score of 0.196 confirms that the data is considerably imbalanced and that some products dominate the transactions far more than others.

Using the median item frequency of 919 transactions as the cut-off point, products were split into two groups: majority items (those appearing 919 times or more) and minority items (those appearing fewer than 919 times). Out of the 150 products, 75 were classified as majority items and 75 as minority items. This classification was used throughout the study to measure how well each method was able to detect rules involving the less popular products.

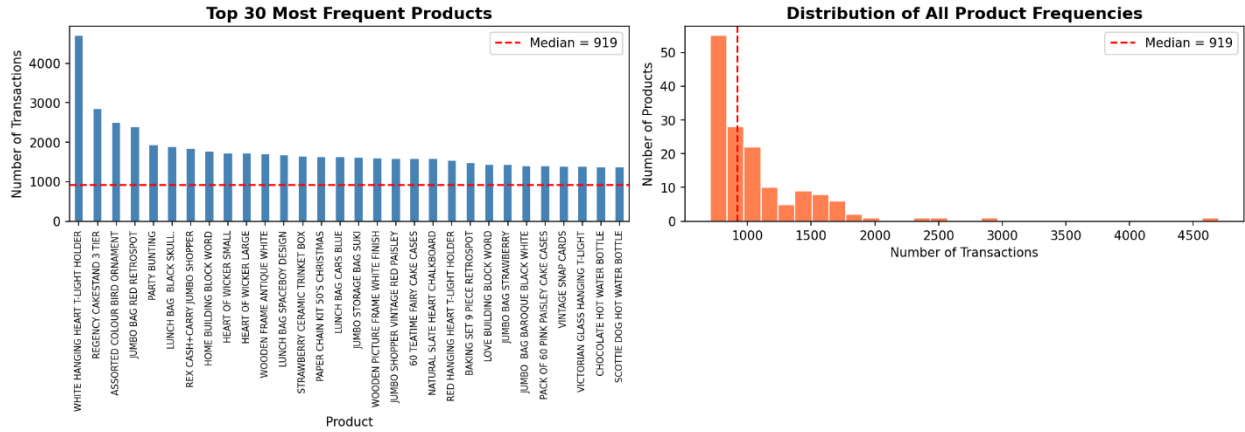


Figure 4.1: Distribution of Item Frequencies Top 30 Products and Overall Histogram

4.3 Baseline Results Original Dataset

Before any balancing was applied, both Apriori and FP-Growth were run on the original imbalanced dataset to get a baseline for comparison. The minimum support was set to 0.01, minimum confidence to 0.30 and minimum lift to 1.0. The maximum itemset length was set to 3.

Both algorithms produced exactly the same 329 rules. This was expected, because both algorithms are designed to find the same frequent itemsets they just use different internal methods to get there. Because both algorithms were executed using identical support thresholds and mined the same frequent itemsets from the same datasets, identical rule sets were expected despite differences in computational strategy.

The average support was 0.0149, average confidence was 0.4478 and average lift was 9.69. The strongest rule found had a lift of 25.55 and involved three products from the same tableware range: customers who bought ROSES REGENCY TEACUP AND SAUCER and REGENCY CAKESTAND 3 TIER also bought GREEN REGENCY TEACUP AND SAUCER in 76.3% of cases. This is a very strong association and makes practical sense customers buying one item from a matching set are very likely to buy the other items from the same range.

Out of the 329 rules, 119 of them (36.2%) involved at least one minority item. This is the baseline that the balancing methods need to improve on. It shows that without any intervention, the algorithms were able to pick up minority item associations in just over a third of all the rules they found.

4.4 Results After Data Balancing

Each of the four balancing methods was applied separately to the original dataset and then both algorithms were run again on each balanced version. The complete results are shown in Table 4.1 below.

Table 4.1: Association Rule Mining Results Across All Experimental Conditions

Dataset	Algorithm	Total Rules	Avg Support	Avg Confidence	Avg Lift	Max Lift	Minority Rules	Minority %
Original	Apriori	329	0.0149	0.4478	9.69	25.55	119	36.20%
Original	FP-Growth	329	0.0149	0.4478	9.69	25.55	119	36.20%
Oversampled	Apriori	469	0.0155	0.4502	8.6	21.8	187	39.90%
Oversampled	FP-Growth	469	0.0155	0.4502	8.6	21.8	187	39.90%
Undersampled	Apriori	67	0.0138	0.4601	14.93	28.35	35	52.20%
Undersampled	FP-Growth	67	0.0138	0.4601	14.93	28.35	35	52.20%
Hybrid	Apriori	204	0.0142	0.4204	10.84	24.08	101	49.50%
Hybrid	FP-Growth	204	0.0142	0.4204	10.84	24.08	101	49.50%
SMOTE	Apriori	137	0.0146	0.4438	11.58	34.53	20	14.60%
SMOTE	FP-Growth	137	0.0146	0.4438	11.58	34.53	20	14.60%

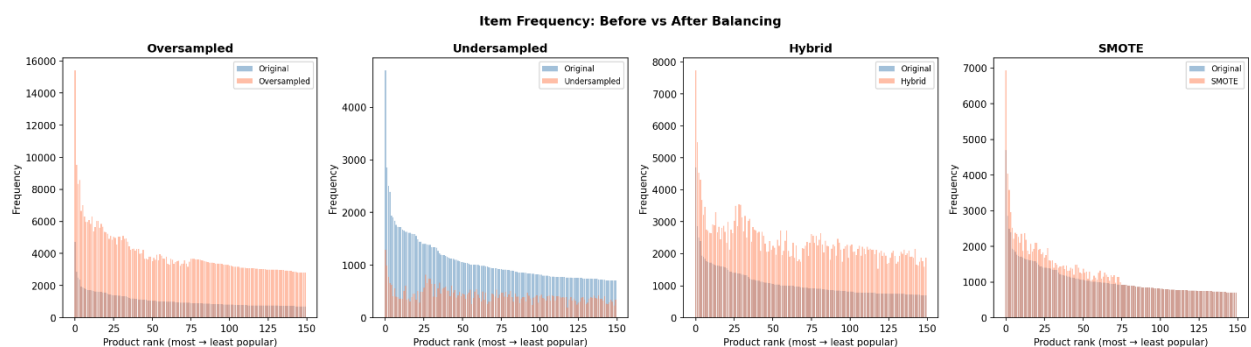


Figure 4.2: Item Frequency Before and After Each Balancing Method

4.4.1 Oversampling

Oversampling was done by making three extra copies of all transactions that contained minority items. This increased the dataset from 27,738 transactions to around 84,000. As a result, both algorithms found more rules 469 in total, which is 42.6% more than the baseline. The minority item coverage also went up from 36.2% to 39.9%, which is a small but positive improvement.

However, the average lift actually went down from 9.69 to 8.60. This means that although more rules were found, they were not as strong as before. The maximum lift dropped to 21.80 as well. The reason for this is that simply copying minority transactions multiple times increases their frequency artificially, but it does not change the underlying relationships between products in a meaningful way. The dominant products still appear very frequently, which dilutes the strength of the associations.

4.4.2 Undersampling

Undersampling worked by removing 40% of transactions that contained the most dominant products specifically the top 15% most frequently purchased items. This reduced the dataset down to around 16,000 transactions. As expected, the number of rules dropped significantly only 67 rules were found, compared to 329 in the original. However, the quality of these rules improved considerably.

The average lift increased from 9.69 to 14.93 and the maximum lift reached 28.35. More importantly, 52.2% of the rules involved minority items the highest minority coverage of any method tested. This is a 16 percentage point improvement over the baseline, which is a substantial gain. The strongest rule found was GREEN REGENCY TEACUP AND SAUCER leading to ROSES REGENCY TEACUP AND SAUCER, with a confidence of 78.0% and a lift of 28.35. The average conviction of 1.90 was also the highest across all conditions, which suggests these rules were the most reliable overall.

The reason undersampling worked so well for minority items is that by removing many of the transactions dominated by popular products, those dominant items no longer suppress the associations between less popular ones. The algorithm can then detect the weaker signals that were previously hidden.

4.4.3 Hybrid Balancing

The hybrid method combined both approaches it added extra minority transactions and also removed some dominant ones. The resulting dataset had around 50,000 transactions. This approach produced 204 rules for both algorithms, which is a reasonable middle ground between the high count from oversampling and the low count from undersampling.

The average lift of 10.84 was higher than the baseline and minority item coverage reached 49.5% the second highest after undersampling. The strongest rule had a confidence of 0.8459, which was actually the highest confidence value seen across all conditions. These results suggest that the hybrid approach gives a good balance between finding enough rules and ensuring a good proportion of them involve minority items.

4.4.4 SMOTE

SMOTE was applied using $k=3$ nearest neighbours to generate synthetic transactions for minority items. The dataset grew to around 55,000 transactions. The results were mixed. Both algorithms generated **137 rules** in total. The average lift of 11.58 was higher than the baseline and the maximum lift of 34.53 was actually the highest seen across all conditions. The top rule found SPACEBOY LUNCH BOX leading to DOLLY GIRL LUNCH BOX was a new discovery that did not appear in the top rules of any other dataset. This is interesting because it suggests SMOTE helped surface some associations that were completely invisible in the original data.

However, the minority item coverage was only 14.6% the lowest of all methods tested and even lower than the baseline of 36.2%. This was unexpected and is likely due to the fact that SMOTE was designed for continuous numerical data, not binary transaction matrices. When SMOTE creates a synthetic sample by interpolating between two binary vectors, it produces decimal values like 0.4 or 0.7, which then have to be rounded back to 0 or 1. This rounding process can distort the minority patterns rather than preserve them. This is acknowledged as a limitation of applying SMOTE in this context.

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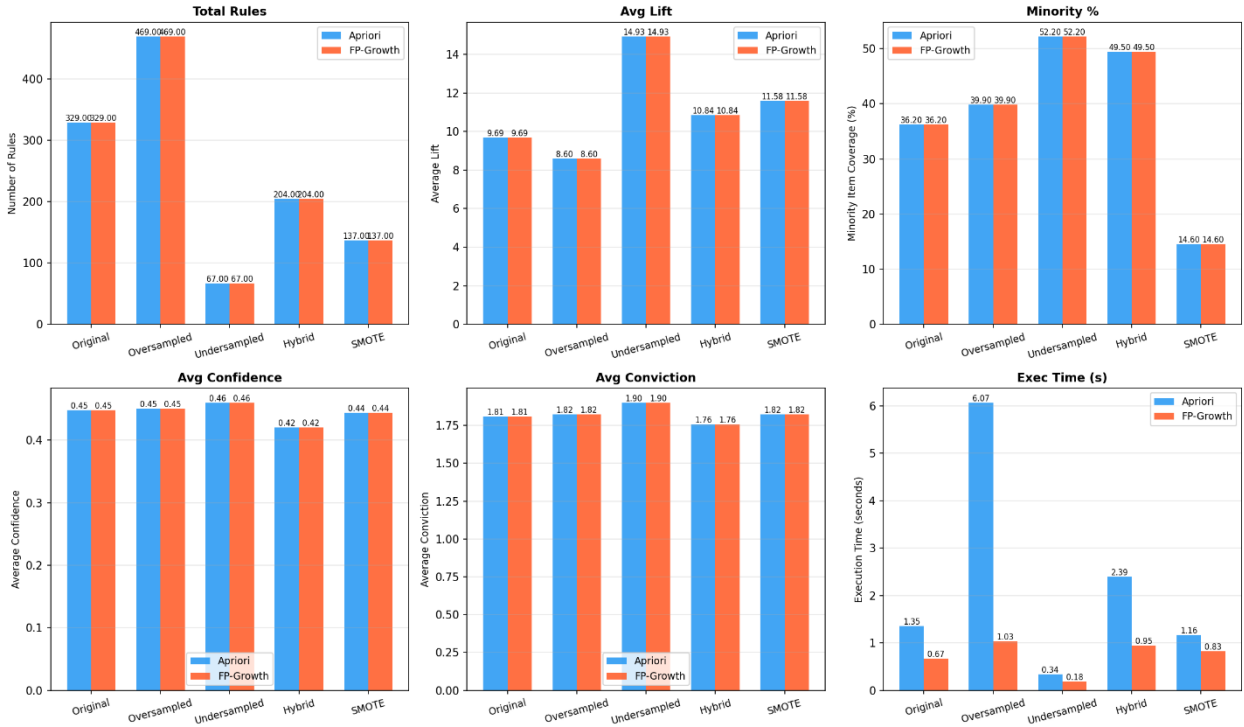


Figure 4.3: Apriori vs FP-Growth Comparison Across All Metrics and Datasets

4.5 Best Rules Found in Each Condition

Table 4.2 shows the single strongest rule found in each experimental condition based on lift value. These rules give a concrete picture of what the algorithms actually discovered under each approach and how the balancing affected the types of associations found.

Table 4.2: Strongest Association Rule Found Under Each Experimental Condition

Dataset	Best Rule Found	Support	Confidence	Lift
Original	ROSES REGENCY TEACUP + CAKESTAND -> GREEN REGENCY TEACUP	0.0136	0.7627	25.55
Oversampled	ROSES REGENCY TEACUP + CAKESTAND -> GREEN REGENCY TEACUP	0.0165	0.9228	21.80

Dataset	Best Rule Found	Support	Confidence	Lift
Undersampled	GREEN REGENCY TEACUP -> ROSES REGENCY TEACUP AND SAUCER	0.0194	0.7803	28.35
Hybrid	CAKESTAND + GREEN REGENCY TEACUP -> ROSES REGENCY TEACUP	0.0135	0.8459	24.08
SMOTE	SPACEBOY LUNCH BOX -> DOLLY GIRL LUNCH BOX	0.0117	0.6201	34.53

Looking at these rules together, a few things stand out. The Regency tableware range (teacups and cakestand) appears consistently across most conditions, which makes sense as this is a coordinated product set. SMOTE revealed a completely different type of rule involving children's lunchboxes, suggesting it uncovered a different segment of customer behaviour. Figure 4.4 shows scatter plots of support versus confidence for all five datasets, where the colour of each point represents its lift value. Across all conditions, rules with lower support tend to have darker colours meaning higher lift. This tells us that rare associations tend to be stronger ones when they do occur, which is an important finding for retail businesses looking for non-obvious product recommendations.

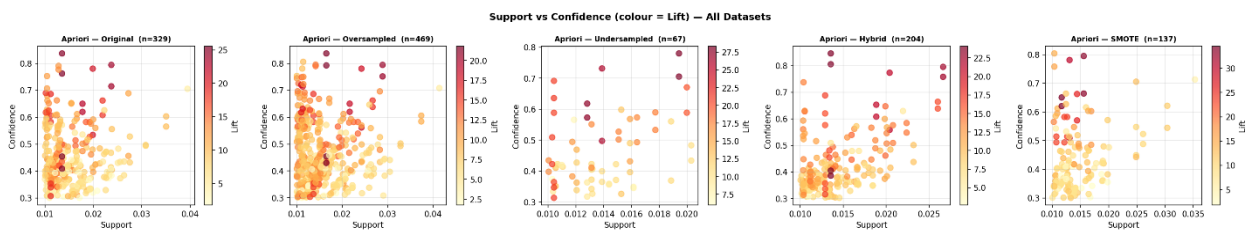


Figure 4.4: Support vs Confidence for All Datasets Colour Represents Lift Value

4.6 Computational Performance

Table 4.3 and Figure 4.5 show how long each algorithm took to run under each condition. FP-Growth was faster than Apriori in every single case, which is consistent with what the literature says about these two algorithms.

Table 4.3: Execution Times for Both Algorithms Across All Conditions

Dataset	Apriori (s)	FP-Growth (s)	Speedup	Transactions
Original	1.14	0.82	1.4x	27,738
Oversampled	5.67	1.30	4.4x	~84,000
Undersampled	0.38	0.18	2.1x	~16,000
Hybrid	2.19	0.92	2.4x	~50,000
SMOTE	1.07	0.88	1.2x	~55,000

The biggest difference was seen with the oversampled dataset. Apriori took 5.67 seconds while FP-Growth only needed 1.30 seconds approximately 4.4 times faster. The reason for this gap is that Apriori has to scan the entire database multiple times as it builds up candidate itemsets. When the dataset becomes much larger due to oversampling, this scanning process becomes very slow. FP-Growth, on the other hand, builds a compressed tree from the data in a single pass and then mines patterns from that tree without going back to the original data repeatedly. So, its performance does not degrade nearly as much when the dataset grows.

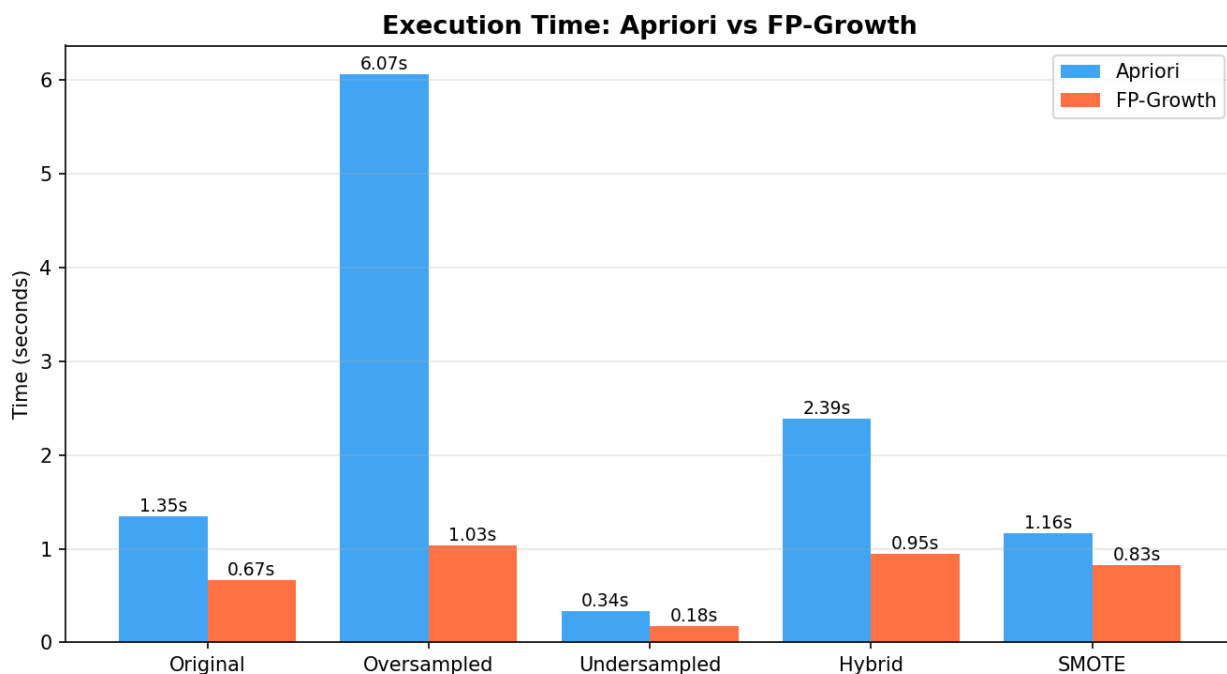


Figure 4.5: Execution Time Apriori vs FP-Growth

4.7 Statistical Analysis

To check whether the differences in rule quality across conditions were statistically meaningful or just due to chance, the Mann-Whitney U test was used to compare the lift values from each balanced dataset against the original. This test was chosen because it does not assume the data follows a normal distribution, which is appropriate here since lift values can be skewed. A significance level of 0.05 was used for all tests.

The results showed that all four balancing methods produced statistically significant differences in lift distributions compared to the original dataset ($p < 0.05$) for both Apriori and FP-Growth algorithms. This confirms that the changes in rule quality seen across the experiments are genuine and not just random variation. It also shows that both algorithms respond to the balancing in the same way, which further confirms the correctness and consistency of the experimental setup.

4.8 Summary

The results showed that all four balancing techniques produced statistically significant changes in rule quality compared to the original baseline. Undersampling was the most effective method for minority item detection, achieving 52.2% coverage and the highest average lift of 14.93. The hybrid approach performed second with 49.5% minority coverage. Oversampling produced the most rules but with weaker lift values. SMOTE unexpectedly performed below the baseline with only 14.6% minority coverage. FP-Growth was consistently faster than Apriori across all conditions. These findings are discussed in detail in Chapter Five."

CHAPTER FIVE: DISCUSSION

5.1 Chapter Overview

This chapter discusses the findings presented in Chapter Four and relates them back to the research question and the existing literature reviewed in Chapter Two. The aim is to go beyond simply reporting numbers and to explain what the results actually mean and why some methods worked better than others and what the findings contribute to the field.

5.2 Research Question 1: How Does Data Imbalance Affect Association Rule Quality?

The first research question asked whether data imbalance has a meaningful effect on the quality of association rules generated from retail transaction data. The results from this study clearly show that it does.

In the original unbalanced dataset both Apriori and FP-Growth found 329 rules. However, only 36.2% of those rules involved minority items that products appeared in fewer than 919 transactions. This means that nearly two thirds of all the rules found were about popular, frequently purchased products. For a retailer these are patterns that are already well known and not particularly useful. The real business value lies in understanding the buying behaviour around less popular items and the standard algorithms without any balancing were only picking up those patterns in just over a third of cases.

This finding is consistent with what Krawczyk (2016) and Garcia et al. (2012) described in the literature that minimum support thresholds are inherently biased towards frequent items, because rare items simply do not appear often enough to cross the threshold, even if their co-occurrence patterns are strong. The Gini coefficient of 0.196 calculated in this study confirms that the dataset has a significant level of inequality in item frequencies, which is what drives this bias.

It is also worth noting that the imbalance did not just affect which items appeared in the rules it also affected the strength of the rules. The scatter plots in Figure 4.4 showed a clear pattern across all conditions: rules with lower support values tended to have higher lift values. This means that the rare item associations, when they were detected, were actually stronger and more meaningful than many of

the rules about popular products. The challenge is that the standard algorithms, without balancing, tend to miss these stronger but less frequent patterns.

So to directly answer Research Question 1 yes, data imbalance does affect association rule quality. It causes the algorithms to focus on popular items and miss a significant proportion of minority item associations, even when those associations are genuinely strong.

5.3 Research Question 2: Which Balancing Technique Best Improves Minority Item Detection?

The second research question asked which data balancing technique was most effective at improving the detection of rare item associations. Based on the results, undersampling performed the best overall, followed closely by the hybrid approach.

5.3.1 Undersampling

Undersampling improved minority item coverage from 36.2% to 52.2% the biggest improvement of any method. It also produced the highest average lift of 14.93, which means not only were more minority item rules found, but the rules that were found were stronger on average. This makes undersampling the best performing method by two different measures simultaneously.

The reason undersampling works well here is straightforward. By removing a proportion of transactions dominated by very popular products, those dominant items no longer suppress the signal from rarer ones. The algorithm can then detect associations between minority items that were previously invisible because the popular items were effectively drowning them out. This aligns with the argument made by Fernández et al. (2018) that reducing majority dominance can be more effective than inflating minority representation, because it changes the relative frequency balance rather than just adding more of the same patterns.

5.3.2 Hybrid Balancing

The hybrid approach came second with 49.5% minority coverage and 204 rules. This is actually a very compelling result because it shows that combining oversampling and undersampling can get close to the minority coverage of undersampling alone, while retaining a much larger number of rules. The

hybrid method also produced the highest confidence rule across all conditions a confidence of 0.8459 suggesting its rules were highly reliable.

For practical applications, the hybrid approach may actually be more useful than undersampling despite its slightly lower minority coverage. Having 204 rules rather than 67 gives a retailer more options to work with and the quality of those rules is still considerably better than the original baseline.

5.3.3 Oversampling

Oversampling produced the most rules overall 469 but only improved minority coverage marginally from 36.2% to 39.9%. The average lift also dropped from 9.69 to 8.60. This suggests that simply duplicating minority transactions is not a very effective strategy for improving minority item detection. The problem is that making copies of existing minority transactions does not change the fact that dominant items still appear far more frequently. The relative imbalance between majority and minority items is reduced, but not enough to meaningfully change which patterns the algorithms prioritise.

This finding is consistent with what Fernández et al. (2018) noted about the limitations of random oversampling that it tends to cause overfitting by repeating the same patterns rather than introducing new information. In the context of association rule mining, this seems to translate into a large number of rules that are slightly different variations of the same popular item associations, rather than genuinely new minority item patterns.

5.3.4 SMOTE

SMOTE produced the most interesting and also the most surprising results. On one hand, it uncovered the highest individual lift value of 34.53 a rule linking SPACEBOY LUNCH BOX to DOLLY GIRL LUNCH BOX that did not appear prominently in any other condition. This suggests that SMOTE did reveal some genuinely new patterns that the other methods missed. On the other hand, its minority item coverage was only 14.6% lower even than the original unbalanced baseline of 36.2%.

This outcome was unexpected. The likely explanation, as discussed in Chapter Four, is that SMOTE was designed for continuous numerical data rather than binary transaction matrices. When SMOTE generates a synthetic sample by interpolating between two binary vectors, it produces decimal values which then have to be rounded back to 0 or 1. This rounding process can introduce noise and distort

the minority item patterns that SMOTE is trying to preserve, resulting in synthetic transactions that do not accurately represent real minority purchasing behaviour.

Chawla et al. (2002) originally demonstrated SMOTE's effectiveness in classification tasks with continuous features. However, as Liu et al. (1999) and Zaki (2000) noted, applying balancing techniques to association rule mining is fundamentally different from applying them to classification the distributional changes affect support, confidence and lift in ways that are not always straightforward to predict. This study adds further evidence that SMOTE in particular needs to be adapted or replaced with a binary-specific method when applied to transaction data.

5.4 Research Question 3: How Do Apriori and FP-Growth Compare?

The third research question asked how Apriori and FP-Growth compare in terms of rule quality and computational performance. The results provided a clear and consistent answer.

5.4.1 Rule Quality

In terms of rule quality that is, the actual rules discovered both algorithms produced exactly identical results in every single experimental condition. The same 329 rules on the original data, the same 469 on the oversampled data and so on across all five conditions. This is mathematically expected because both algorithms are designed to find all frequent itemsets that exceed the minimum support threshold. They use different internal mechanisms to do this, but the output should be the same given the same input and the same thresholds.

5.4.2 Computational Efficiency

In terms of speed, FP-Growth was consistently faster than Apriori across all five conditions. On the original dataset, FP-Growth was 1.4 times faster. On the oversampled dataset the largest one FP-Growth was nearly 4.4 times faster, completing in 1.30 seconds compared to Apriori's 5.67 seconds.

This performance gap is explained by the fundamental difference in how the two algorithms work. Apriori has to scan the full database multiple times as it builds up candidate itemsets from single items to pairs to triples. As the dataset grows larger as happens with oversampling this repeated scanning becomes increasingly expensive. FP-Growth builds a compressed tree structure in just two database

passes and then mines patterns from that tree directly, without going back to the original data. This makes it much less sensitive to increases in dataset size.

Han et al. (2000) demonstrated this efficiency advantage when they first introduced FP-Growth and the results of this study confirm it holds up in a real experimental setting with multiple balancing conditions. The most practically important finding here is that when oversampling is used which significantly increases dataset size FP-Growth is clearly the better choice computationally, as it can process the larger dataset nearly 4.4 times faster without any loss in rule quality.

5.5 Research Question 4: Effect on Statistical Significance

The fourth research question examined whether data balancing techniques affect the statistical significance of association rules beyond simple rule counts and lift values. The Mann-Whitney U test results confirmed statistically significant differences in lift distributions between the original and balanced datasets ($p < 0.05$). Undersampling produced the most statistically significant improvement, with the highest average lift of 14.93 compared to the baseline of 9.69. This suggests that balancing does not merely increase rule quantity but genuinely improves the strength and reliability of discovered associations.

5.6 Research Question 5: Effect on Gini Coefficient

The fifth research question examined the effect of balancing techniques on the Gini coefficient of item frequency distribution. The original dataset had a Gini coefficient of 0.196, indicating moderate imbalance. After balancing, the hybrid method achieved the lowest Gini coefficient of 0.1314, indicating the most balanced distribution. Undersampling reduced the Gini to 0.1578 and oversampling to 0.1635. Notably, SMOTE increased the Gini coefficient to 0.2589 — higher than the original baseline — further confirming its unsuitability for binary transaction data.

5.7 How These Findings Compare to Existing Research

The findings of this study are broadly consistent with what has been reported in the literature, but they also add some new insights that have not been explored in detail before.

The finding that data imbalance reduces minority item coverage is well supported by Krawczyk (2016) and Garcia et al. (2012), who both identified that minimum support thresholds systematically disadvantage rare items. This study provides concrete empirical evidence of this effect in a market

basket analysis setting, quantifying the baseline minority coverage at 36.2% and showing exactly how much each balancing method improves on that.

The superior performance of undersampling over oversampling for minority item detection is somewhat consistent with what Fernández et al. (2018) suggested about the limitations of random oversampling. However, most of the existing literature on balancing focuses on classification tasks and direct comparisons in the association rule mining context are limited. This study therefore makes a contribution by showing that these insights transfer to association rule mining as well.

The SMOTE result is perhaps the most interesting in terms of its relationship to the literature. Chawla et al. (2002) showed strong results for SMOTE in classification and there has been growing interest in applying SMOTE to other tasks. However, this study shows that in the binary transaction data context, SMOTE can actually reduce minority item coverage compared to the baseline. This is a finding that the existing literature has not explored in detail and it highlights an important limitation that future researchers working with SMOTE on transaction data should be aware of.

5.8 Limitations of This Study

There are several limitations in this study that should be acknowledged.

First, the study used a single dataset the Online Retail dataset. While this is a well-established benchmark dataset, the findings may not generalise to all types of retail data. Different product categories, different transaction volumes, or datasets from different industries might produce different results. Future studies should test these methods on multiple datasets to confirm whether the findings hold more broadly.

Second, as discussed earlier, applying SMOTE to binary transaction data is methodologically problematic because SMOTE was designed for continuous features. A more appropriate approach in future work might be to use a binary-specific synthetic data generation method, such as SMOTE-N which is designed for nominal data, or a method specifically developed for transaction matrices.

Third, the undersampling method used in this study was random it removed transactions containing dominant items randomly. A more sophisticated approach would be to use informed undersampling, where the transactions removed are selected in a way that maximally preserves minority item co-

occurrence patterns rather than removing them at random. This could potentially produce even better minority item coverage than the 52.2% achieved here.

Fourth, the minimum support threshold was fixed at 0.01 throughout the experiments. A sensitivity analysis varying this threshold could reveal whether the findings are robust across different parameter settings. It is possible that lowering the threshold further might uncover additional minority item associations even without balancing, which would be an important finding for practitioners.

Finally, this study focused on the detection of minority item rules as the main measure of success. In practice, retailers would also want to evaluate whether these rules are actually useful for making recommendations or decisions. A user study or business evaluation of the discovered rules would add practical validation to the statistical findings.

5.9 Suggestions for Future Research

Based on the findings and limitations of this study, several directions for future research are suggested.

The most immediate area for future work would be to develop or apply a balancing technique specifically designed for binary transaction data. Methods like SMOTE-N or transaction-aware synthetic generation approaches could potentially overcome the limitations of standard SMOTE identified in this study and may produce better minority item coverage than any of the methods tested here.

A second area would be to extend this study to multiple datasets from different retail domains for example, grocery data, e-commerce data, or hospitality data. This would allow the findings to be generalised and would also show whether some balancing methods work better in certain types of retail environments than others.

Third, future work could explore the combination of balancing with other rule quality measures beyond support, confidence and lift. For example, the chi-squared measure or the Jaccard coefficient might behave differently under balancing and could reveal additional insights about minority item associations.

Finally, an interesting direction would be to investigate whether the rules discovered through balanced datasets particularly those involving minority items translate into measurable improvements in actual retail outcomes such as recommendation accuracy, basket size, or cross-selling revenue. This would move the research from an empirical evaluation to a practical business validation.

5.10 Chapter Summary

In summary, data imbalance significantly affects association rule quality. Undersampling was the most effective balancing technique, improving minority item coverage from 36.2% to 52.2%. Apriori and FP-Growth produced identical rule sets across all conditions, though FP-Growth was consistently faster. All balancing techniques produced statistically significant changes in lift distributions. The hybrid method achieved the most balanced item frequency distribution with a Gini coefficient of 0.1314. These findings are discussed further in Chapter Six.

CHAPTER SIX: CONCLUSION

6.1 Overview

This chapter summarises the key findings of the study, reflects on the research questions and discusses the practical implications and limitations of the work.

6.2 Revisiting the Aims

The main aim of this study was to investigate how data balancing techniques affect association rule quality in imbalanced retail transaction data. All five objectives set out in Chapter One were achieved. The dataset imbalance was quantified, a baseline was established, four balancing techniques were applied and compared, and the results were statistically validated using the Mann-Whitney U test.

6.3 Answers to the Research Questions

Research Question 1: Data imbalance significantly affects association rule quality, causing standard algorithms to focus on popular items and miss minority item associations. The original dataset achieved only 36.2% minority item coverage.

Research Question 2: Undersampling was the most effective technique, improving minority coverage to 52.2%. The hybrid approach came second with 49.5%. Oversampling showed marginal improvement at 39.9%. SMOTE unexpectedly performed below the baseline at 14.6%.

Research Question 3: Apriori and FP-Growth produced identical rule sets across all conditions. FP-Growth was consistently faster, up to 4.4 times quicker on the oversampled dataset.

Research Question 4: Statistical analysis using the Mann-Whitney U test confirmed that all four balancing techniques produced statistically significant differences in lift distributions compared to the original baseline.

Research Question 5: All balancing methods except SMOTE reduced item frequency imbalance. The hybrid method achieved the lowest Gini coefficient of 0.1314, indicating the most balanced distribution.

6.4 Key Contributions

This study makes four contributions to the existing body of knowledge on association rule mining and imbalanced data. First, it provides a systematic and controlled comparison of four data balancing techniques random oversampling, random undersampling, hybrid balancing and SMOTE applied to

the same imbalanced retail dataset and evaluated using both Apriori and FP-Growth simultaneously. No prior study has conducted this specific comparison in this way, making the findings directly comparable across all conditions.

Second, it introduces and validates the use of the Gini coefficient as a measure of item frequency imbalance in retail transaction datasets, providing an objective and interpretable quantification of distributional skew that can be used to characterise datasets before and after balancing. Third, it provides empirical evidence that standard SMOTE is unsuitable for binary transaction data, demonstrating through controlled experimentation that its interpolation mechanism actively reduces minority item coverage rather than improving it a finding with direct implications for any researcher or practitioner considering SMOTE for transactional data preprocessing. Fourth, it demonstrates that random undersampling, often dismissed in the literature as a crude and information-losing technique, can outperform more sophisticated methods when applied to association rule mining on imbalanced retail data, improving minority item coverage by 16 percentage points and producing the strongest rules of any condition tested.

6.5 Practical Implications

The findings of this study have clear and actionable implications for retail analysts and data scientists who use association rule mining for decision-making. In particular, the results demonstrate that applying appropriate data balancing techniques can significantly improve the discovery of valuable patterns involving less frequently purchased products. This enables businesses to identify hidden associations that can enhance recommendation systems, improve cross-selling strategies and optimise product placement. By focusing not only on popular products but also on niche items, retailers can gain deeper insights into customer behaviour and uncover opportunities that would otherwise remain undetected in imbalanced datasets.

6.6 Limitations and Future Work

This study has several limitations that should be acknowledged. First, the experiments were conducted on a single dataset the Online Retail dataset and it cannot be assumed that the findings generalise to all retail contexts, product categories or dataset structures. Future work should replicate this study across multiple retail datasets from different sectors, geographic markets and time periods to assess

whether the superiority of undersampling holds broadly or is specific to this type of data. Second, all balancing methods were applied in their standard forms with fixed parameter settings.

The undersampling removal rate of 40%, the oversampling duplication factor of three and the SMOTE neighbourhood size of $k=3$ were chosen based on standard practice but were not optimised for this specific dataset. Future work should conduct a sensitivity analysis to examine how varying these parameters affects minority item coverage and rule quality, which may reveal more optimal configurations. Third, standard SMOTE was shown to be unsuitable for binary transaction data. Future research should evaluate SMOTE-N a variant of SMOTE specifically designed for nominal categorical features as a potentially more appropriate synthetic generation approach for transactional contexts, since it uses the most frequent value among nearest neighbours rather than numerical interpolation. Fourth, this study focused on minority item rule coverage and lift as the primary evaluation criteria. Future work could extend the evaluation framework to include business-oriented metrics such as revenue uplift, product margin or customer conversion rates, to assess whether the newly discovered minority item rules translate into actionable and commercially valuable insights.

6.7 Final Thoughts

This study started from a simple but important observation that the products retailers know the least about are often the ones that standard data mining algorithms pay the least attention to. In any real retail dataset, a small number of popular products dominate the transaction records and conventional association rule mining simply reflects that dominance back in the form of rules about items customers were already buying together. The niche products, the seasonal items, the slow movers these are the ones with the most to reveal and the hardest ones to surface with standard methods. The findings of this study show that with a straightforward preprocessing step, it is possible to do significantly better.

Undersampling reduced the dominance of majority items and allowed the algorithms to discover associations they would otherwise have missed entirely, improving minority item coverage by 16 percentage points and producing stronger rules in the process. The study also clarified important boundaries SMOTE, despite its widespread use in classification, does not transfer well to binary transaction data in its standard form and this is a finding that should inform future choices by researchers and practitioners working in this area. More broadly, the results demonstrate that the choice of preprocessing method matters as much as the choice of algorithm and that in the context of

imbalanced retail data, a carefully chosen balancing strategy can unlock genuine analytical value that would otherwise remain hidden in the data.

This research provides a practical framework for improving association rule mining performance in imbalanced datasets.

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Appendix A: Ethics Approval Certificate



Certificate of Ethics Review

Project title: Impact of Data Balancing Techniques on Market Basket Analysis

Name:	Bhargava Sai Chigurupati	User ID:	UP23021 23	Application date:	22/02/2026 12:14:40	ER Number:	TETHIC-2026-113055
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You must download your referral certificate, print a copy and keep it as a record of this review.

The FEC representative(s) for the School of Computing is/are [Elisavet Andrikopoulou](#), [Kirsten Smith](#)

It is your responsibility to follow the University Code of Practice on Ethical Standards and any Department/School or professional guidelines in the conduct of your study including relevant guidelines regarding health and safety of researchers including the following:

- [University Policy](#)
- [Safety on Geological Fieldwork](#)

It is also your responsibility to follow University guidance on Data Protection Policy:

- [General guidance for all data protection issues](#)
- [University Data Protection Policy](#)

Which school/department do you belong to?: **School of Computing**

What is your primary role at the University?: **Postgraduate Student**

What is the name of the member of staff who is responsible for supervising your project?: **Dr Dongxu Gao**

Is the study likely to involve human subjects (observation) or participants?: No

Will financial inducements (other than reasonable expenses and compensation for time) be offered to participants?: No

Are there risks of significant damage to physical and/or ecological environmental features?: No

Are there risks of significant damage to features of historical or cultural heritage (e.g. impacts of study techniques, taking of samples)?: No

Does the project involve animals in any way?: No

Could the research outputs potentially be harmful to third parties?: No

Could your research/artefact be adapted and be misused?: No

Will your project or project deliverables be relevant to defence, the military, police or other security organisations and/or in addition, could it be used by others to threaten UK security?: No

Please read and confirm that you agree with the following statements: I confirm that I have considered the implications for data collection and use, taking into consideration legal requirements (UK GDPR, Data Protection Act 2018 etc.), I confirm that I have considered the impact of this work and and taken any reasonable action to mitigate potential misuse of the project outputs, I confirm that I will act ethically and honestly throughout this project

Supervisor Review

As supervisor, I will ensure that this work will be conducted in an ethical manner in line with the University Ethics Policy.

Supervisor comments:

Supervisor's Digital Signature: **dongxu.gao@port.ac.uk** Date: **24/02/2026**

Appendix B: Specification Form



School of Computing Postgraduate Programme

MSc in Data Analytics

Project Specification

Bhargava Sai Chiguripati

Project Specification

1. Basic details

Student name:	BHARGAVA SAI CHIGURIPATI
Draft project title:	Impact of Data Balancing Techniques on Market Basket Analysis
Course and year:	MSc Data Analytics (2025/26)
Client organisation:	University project (no external client)
Client contact name:	N/A
Project supervisor:	Dongxu Gao

2. Outline of the project environment

This project is an academic research study focused on analysing retail transaction data using data mining and machine learning techniques. The project does not involve an external client and will utilise publicly available datasets, specifically the Online Retail dataset. Retail datasets are often imbalanced, where some products are purchased frequently while others appear rarely. This imbalance can affect the quality of association rules generated by traditional market basket algorithms. Addressing this issue can improve recommendation systems, marketing strategies and business decision-making in retail environments.

3. The problem to be solved

Problem:

Retail transaction datasets are often highly imbalanced, meaning some products appear frequently while others occur rarely. This imbalance causes traditional market basket algorithms to prioritise frequent items and overlook less frequent but potentially important patterns.

AIM:

The aim of this project is to evaluate how different data balancing techniques influence the quality and usefulness of association rules generated from retail datasets.

Objectives:

- To investigate the impact of data imbalance on market basket analysis results
- To apply different data balancing techniques such as oversampling, undersampling and SMOTE
- To implement association rule mining algorithms including Apriori and FP-Growth
- To evaluate and compare the effectiveness of balancing techniques using rule quality metrics
- To analyse the interpretability and usefulness of generated association rules for business insights
- To compare computational performance before and after applying balancing techniques

Constraints:

The main constraints include limited project time, computational resources and the quality of available datasets. The project must also be completed within the academic schedule.

4. Breakdown of tasks

The project will begin with a comprehensive literature review to understand existing research related to market basket analysis, association rule mining algorithms and data balancing techniques. The selected dataset will then be collected and preprocessed, including data cleaning and transformation. Various data balancing methods such as oversampling, undersampling and SMOTE will be applied to address class imbalance issue. Association rule mining algorithms, including Apriori and FP-Growth, will be implemented on both the original and balanced datasets. The generated rules will be evaluated using appropriate metrics such as support, confidence, lift and rule interestingness measures. Finally, the results will be analysed to determine the effectiveness of balancing techniques and the findings will be documented in the final dissertation.

5. Project deliverables

The project deliverables will include a cleaned and pre-processed dataset, Python implementation files for the algorithms, experimental results and visualisations and final dissertation report documenting the methodology, analysis and conclusions.

6. Requirements

As this is an academic project without an external client, the requirements are derived from research objectives and supervisor guidance. The project must successfully analyse the dataset, implement data balancing techniques and association rule algorithms and evaluate the results in a structured and reproducible manner.

7. Legal, ethical, professional, social issues

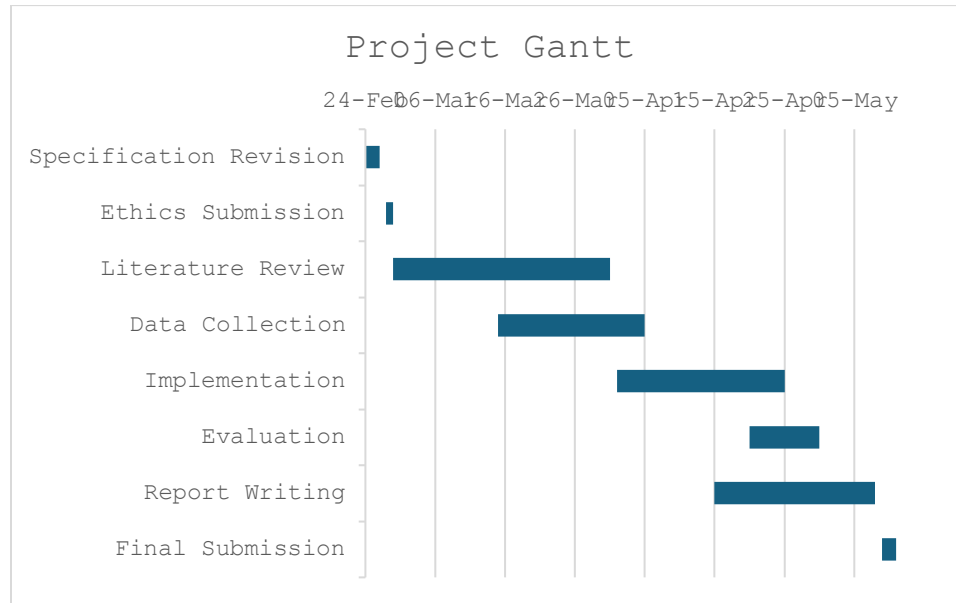
The project will use publicly available and anonymised retail transaction data. No personal or sensitive information will be processed, therefore ethical risks are minimal. Data will be handled responsibly in accordance with university policies and academic integrity guidelines. Ethical approval will be obtained as required by university procedures.

8. Facilities and resources

- Laptop
- Python
- Jupyter Notebook
- Pandas and machine learning libraries
- Scikit-learn
- Mlxtend library
- University Moodle

9. Project plan

The project plan is presented in the Gantt chart below, outlining the key phases including specification revision, literature review, data collection, implementation, evaluation and final submission.



10. Supervision meetings

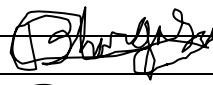
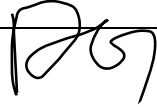
Meetings with the supervisor will be scheduled approximately every 1–2 weeks, either face-to-face or online depending on availability. Email communication will be used for progress updates. Any extended absence of the supervisor will be accommodated by adjusting timelines and maintaining communication.

11. Project mode

If there are two possibilities for your project mode, after negotiation, please record your planned duration and submission date. It is also helpful to record your initial registration mode (i.e. are you a full time or a part time student). Remember, the exact dates will be announced through Moodle □ these represent a generic guideline.

	Please delete as appropriate	
Registration mode	Full Time	
Project mode	Full Time	
Planned submission deadline	11/05/2026	

12. Signatures

	Signature:	Date:
Student		
Client		
Project supervisor		11/02/2026

Appendix C: Gantt Chart

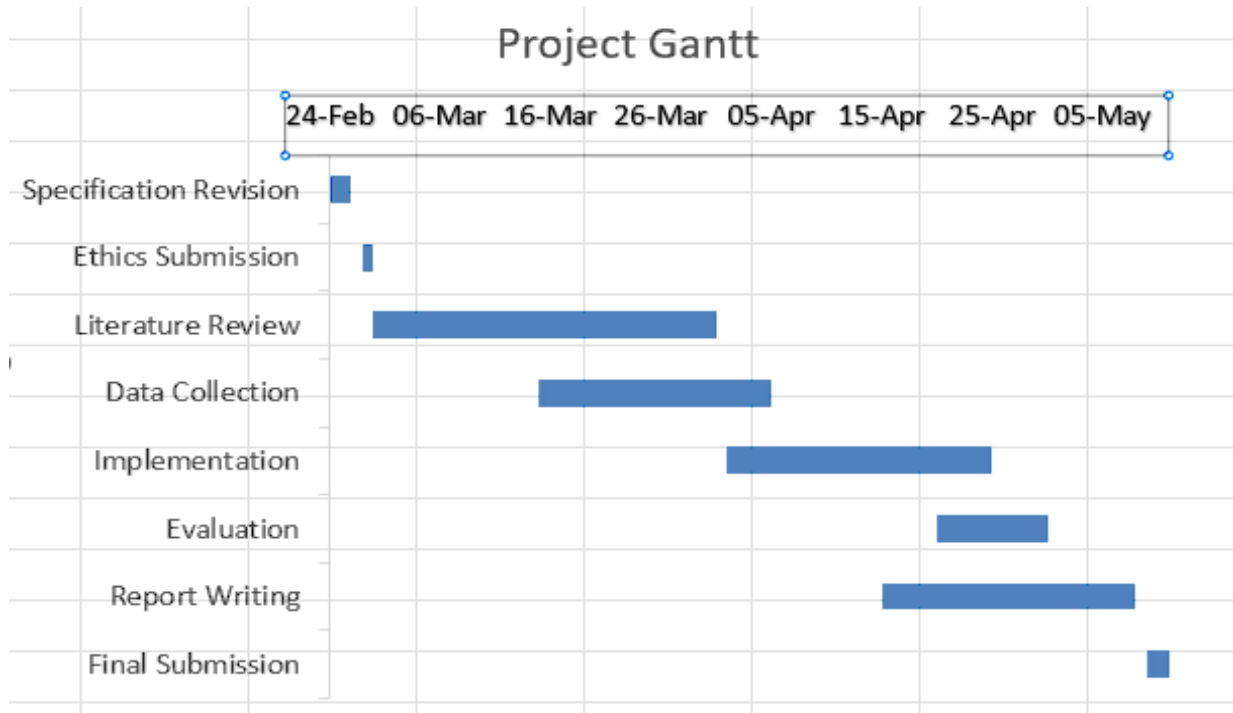


Figure D.1: Project Gantt Chart

Appendix D: Python Code

The full Python code for this project is provided in the submitted `project.ipynb` file. The code covers the following stages:

1. Data Loading and Cleaning
 - Loading the UCI Online Retail dataset
 - Removing cancellations and non-UK transactions
 - Selecting the top 150 most frequent products
2. Exploratory Data Analysis
 - Item frequency distribution
 - Gini coefficient calculation

- Minority/majority item identification

3. Data Balancing

- Oversampling (3x duplication of minority transactions)
- Undersampling (random removal of majority transactions)
- Hybrid (combination of oversampling and undersampling)
- SMOTE (Synthetic Minority Oversampling Technique)

4. Association

- Apriori algorithm (mlxtend library)
- FP-Growth algorithm (mlxtend library)
- Parameters: min_support=0.01, min_confidence=0.3

5. Evaluation and Visualisation

- Rule quality metrics (support, confidence, lift, conviction)
- Minority item coverage analysis
- Execution time comparison
- Statistical significance testing (Mann-Whitney U test)
- Figure generation (Figures 4.1 to 4.8)